

VIVEK VENKATESHPRASAD

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Work Experience - 2.5 years

Software Engineer

Cisco Systems Inc.

Feb 2021 – Aug 2023

Bengaluru, Karnataka, India

- Engineered and deployed 6+ advanced tools and solutions for Cisco's Business Critical Services 3.0 SaaS offering, delivering operational insights into network devices and generating over \$7 million in revenue.
- Identified and resolved performance bottlenecks in distributed systems by implementing techniques like multi-threading, data sharding, and indexing, achieving a 73% reduction in processing time.
- Worked with data scientists to identify key configuration themes across devices using ML/AI models (LSTM, BERT, LLM), resulting in a 30% reduction in customer tickets by recommending optimal configurations.
- Developed optimal distributed training pipelines for machine learning models, leveraging GPU-accelerated cloud solutions with OpenStack, PyTorch, and S3, reducing model training time for 1M+ network device configurations from 8 hours to 1 hour.
- Developed an automated association rule mining and OS-specific best practices parsing engine using Java (Apache POI) and Python (openpyxl, RegEx, BeautifulSoup) to extract structured data from a large corpus.
- Developed microservices for customer device metrics and forecasting, focusing on job scheduling and resource management. Leveraged FastAPI/Spring Boot, OpenShift, MongoDB Atlas and AWS to enhance deployment reliability and scalability to support over 100 customers managing more than 20,000 devices.
- Streamlined CI/CD pipelines for 5+ services by automating build, test, and deployment workflows using Docker, Jenkins, SonarQube, and OpenShift(Kubernetes), saving 14 hours of manual deployment time each week.
- Redesigned UI components using JavaScript to create interactive dashboards for historical data trend analysis and tested the functionality using JUnit and pytest.

Projects

Patronus - An app to help people with dementia | *React Native, Flask, LangChain, Qdrant, Gemini, RAG*

2024

- Co-developed an award-winning app at Cerebral Beach 2024 (L.A.'s largest Gen AI hackathon). The app, designed as a personal assistant and memory bank for individuals with dementia, received top honors, including "Best Zero to One Startup", "Most Responsible Use of AI" and "Most Innovative Project" along with a cash prize.
- Implemented a RAG model powered by the Gemini LLM on top of a Qdrant vector database to store personalized user information. Integrated Twelve Labs' multimodal AI API to enable video querying via text and images. Designed a rich, interactive UI using React Native and developed a robust backend hosted on a Flask server.
- <https://devpost.com/software/patronus-a-spell-against-dementia>

Movie plot generation using predictive text | *Large language modelling(LLM), Pytorch, Sentiment Analysis, RNN*

2024

- Designed and developed a movie plot generation application using predictive text as part of my AI class. The application generates creative movie ideas and predicts potential box office success based on various contributing factors.
- Implemented using a RNN-based language model in PyTorch. Trained the model on the IMDB top movies dataset, incorporating sentiment analysis to enhance plot coherence and creativity. Fine-tuned hyperparameters and optimized training for improved performance, generating compelling and contextually relevant movie ideas.

P.I.G.L.E.T | *React Native, Javascript, Google Firestore/Firebase, Google Cloud Platform*

2021

- "Platform for Issues and Grievances Laid out by Emerging Technologies" is a progressive web application designed for real-time reporting of civic issues to the appropriate authorities.
- Created using React Native for the front-end and Google Firestore/Firebase for data storage and management. Integrated Google Cloud Platform for scalable backend services and implemented real-time notifications to ensure prompt issue resolution by responsible authorities.

Education

University of California, Riverside

Master of Science in Computer Science - GPA: 3.95/4

Sep 2023 – Dec 2024

Riverside, California

B.M.S College of Engineering

Bachelor of Science in Information Science - GPA: 3.6/4

Sep 2017 – Aug 2021

Bengaluru, Karnataka, India

Expertise

Programming Languages: Python, C, C++, Java, HTML5/CSS, SQL, NOSQL/MQL

Technologies/Frameworks: Flask, FastAPI, numpy/pandas, OpenCV, Pytorch, Tensorflow, AWS, GCP, Git, Selenium

Databases: MongoDB/Mongo Atlas, MySQL, DynamoDB, Qdrant